

Where the Bank Quietly Loses Its Customers

Knowledge Discovery Report — Bank Customer Churn

Group 5 · Data Mining · KDD Methodology (Phases 1–5)

10,000 retail-banking customers · France, Germany, Spain · June 2026

1. Executive Summary

One in five customers leaves this bank. That fact sits on the surface of the raw data and is not, by itself, knowledge. Over fourteen weeks we treated the customer file as a discovery exercise — profiling segments, mining co-occurrence rules, and auditing anomalies — and found four things the raw columns do not show.

First, churn has an age and depth structure. Customers aged 46–60 leave at 51%, against 14% for everyone else. When that age band combines with inactivity and a single product, churn reaches 77.3% — nearly four times the bank-wide rate. No single attribute reveals this; it is the interaction that carries the risk.

Second, Germany is a market-level problem. German customers churn at 32.4%, double France (16.2%) and Spain (16.7%). German customers aged 46–60 churn at 67.3% even before considering activity or products, and Germany supplies 46.3% of all high-balance churners.

Third, engagement beats balance as a loyalty signal. The bank's most loyal segment holds the least money (average £6,214) but nearly two products per customer. The wealthiest segments (£115,000–122,000) hold about 1.3 products and churn at 23–24%. Customers with one product churn at 27.7%; with two, at 7.6%.

Fourth, high-value money leaves quietly. Structurally unusual customers — flagged independently by two different detection families — churn at 49–52%. Among them sit 592 high-balance churners averaging £149,755 each, the most expensive losses in the book.

Every finding above is actionable, and none of them is visible in a one-column summary of the raw file. Section 9 turns them into five concrete retention actions.

2. The Question We Were Asked

The brief for this project was explicitly not to build a churn-prediction machine. It was to answer one question: ***“What did we discover that was not already obvious from the raw data?”*** Accuracy scores reward a model for repeating the data back to us; discovery requires explaining the patterns so that a branch manager could act on them. This report is therefore

written in business language. The technical record — every parameter, test and validation — lives in the project notebook, and an interactive dashboard accompanies this report for exploration.

3. What We Worked With

The dataset describes 10,000 retail-banking customers across three markets: France (5,014 customers), Germany (2,509) and Spain (2,477). For each customer we know their age, credit score, account balance, salary estimate, years with the bank, number of products held, credit-card status, activity status, and whether they ultimately closed their account. The data is a snapshot, not a time series: 2,037 customers (20.4%) are marked as having left.

The file arrived unusually clean — no missing values and no duplicates — so our preparation work focused on making the data minable rather than repairing it: identifiers were removed, continuous values were scaled for the segmentation algorithms, and age, balance, credit score, tenure and salary were translated into named business bands (for example “age 46–60” or “balance £50–125K”) so that discovered rules would read as sentences rather than numbers.

4. How We Mined It

Segmentation. Three different clustering algorithms (K-Means, hierarchical, DBSCAN) grouped customers by behavioural and financial similarity without being told who churned. Churn was used only afterwards, as a lens to interpret the groups found.

Rule mining. The Apriori algorithm searched all combinations of customer attributes for profiles that co-occur with churn far more often than chance. Each rule is reported with its share of the customer base (support), the churn rate inside the profile (confidence), and how many times the baseline rate that is (lift).

Anomaly audit. Four independent detectors — two statistical (IQR, Z-score) and two structural (Isolation Forest, DBSCAN noise) — flagged unusual customers. Every flagged record was classified as a data error, a rare-but-legitimate case, or a churn risk signal, so that anomalies are acted on rather than deleted.

Presentation. All findings feed an interactive dashboard (cluster maps, a rule network, outlier plots and distributions) whose every control responds well under a tenth of a second.

5. The Four Discoveries

Discovery 1 — The mid-life churn cliff

Plotting churn against age shows a cliff, not a slope. Customers aged 46–60 — 1,647 people, one-sixth of the book — churn at 51.1%, while everyone else churns at 14.3%. The rulebook sharpens this: inactive 46–60-year-olds holding a single product churn at 77.3% (lift 3.8). Inactivity alone is a much weaker signal; it is the combination with mid-life age and shallow product holdings that makes departure almost certain. Customers over 60, by contrast, are not a high-risk group — the danger zone is specifically the 46–60 band, the years when people consolidate their finances ahead of retirement and quietly choose which bank keeps the money.

Discovery 2 — Germany is a structural problem

Germany's churn rate of 32.4% is double the other two markets. Crucially, the German effect survives every slicing we tried: German customers aged 46–60 churn at 67.3% with no other condition attached, inactive single-product Germans churn at 52.1% (2.6 times baseline), and two of our three customer personas are Germany-heavy with elevated churn. When one market misbehaves across segments, products and age bands simultaneously, the cause is unlikely to be the customers. It points at the offer: pricing, service or product fit in the German operation.

Discovery 3 — Engagement beats balance as a loyalty signal

Ranking customers by deposits would put the loyal ones last. Our most loyal persona holds an average balance of just £6,214 — but 1.88 products per customer, and churns at only 15.7%. The two wealthy personas (£115,296 and £121,520 average balances) hold roughly 1.3 products and churn at 23.6% and 22.8%. Product depth is the mechanism: one-product customers churn at 27.7%, two-product customers at 7.6%. A second product is not extra revenue so much as a retention anchor. Balance measures how much money a customer parks; products measure how much of their financial life they have moved in.

Discovery 4 — High-value money leaves quietly

Two structurally different detectors — one tree-based (Isolation Forest), one density-based (DBSCAN) — independently flagged overlapping sets of “strange” customers (384 records in common), and both sets churn at roughly two and a half times baseline (49.0% and 52.2%). Inside the flagged population sit the most expensive losses: 592 churners with balances above the 75th percentile, averaging £149,755 each — and 46.3% of them are German. Because the dataset is a snapshot, we cannot watch a balance fall before closure; but the pattern of unusually-shaped, high-balance accounts that subsequently closed is the signature such departures would leave.

6. The Three Customer Personas

Clustering found three recognisable customers. The groups overlap at the edges (they are working personas, not walls), but each has a distinct centre of gravity and a clear business meaning.

Persona	Size	Churn	Avg balance	Products	Germany
C0 — Low-Balance Multi-Product Loyalists (France/Spain)	3,735 (37%)	15.7%	£6,214	1.88	2.2%
C1 — Germany-Heavy Single-Product Watchlist (newer)	3,161 (32%)	23.6%	£115,296	1.32	37.7%
C2 — High-Balance Germany-Heavy Passive Holders (long tenure)	3,104 (31%)	22.8%	£121,520	1.33	39.8%

C1 and C2 are the money: two-thirds of deposits sit with customers who hold barely more than one product and churn above baseline. C1 is newer (2.3 years average tenure) and most exposed; C2 has been with the bank 7.5 years but is the least active group (48.9% active). C0

proves the bank *can* create loyalty — it simply has not transferred that recipe to its wealthy customers.

7. The Churn Rulebook

The ten strongest discovered rules, in plain words. “Churn rate” is the observed rate inside the profile (bank-wide baseline: 20.4%); “lift” is how many times the baseline that is; “share” is the proportion of all 10,000 customers who match the profile and churned.

Customer profile	Share	Churn rate	Lift	Read
Inactive + age 46–60 + single product	4.05%	77.3%	3.79	Call first
Inactive + age 46–60	5.38%	68.4%	3.36	Age × inactivity
Inactive + age 46–60 + has credit card	3.78%	67.6%	3.32	Card ≠ anchor
Age 46–60 + Germany	3.38%	67.3%	3.31	Market issue
Age 46–60 + female + single product	3.23%	66.5%	3.26	Hidden cross-tab
Age 46–60 + single product	6.06%	60.9%	2.99	Cross-sell gap
Age 46–60 + has card + single product	4.18%	59.9%	2.94	Card-only = shallow
Age 46–60 + female	4.71%	57.3%	2.81	Watch segment
Age 46–60 + balance £50–125K	3.67%	56.2%	2.76	Loses the middle
Inactive + Germany + single product	3.75%	52.1%	2.56	Confirms brief

Two readings deserve emphasis. Holding a credit card does *not* protect a shallow relationship — card-holding seniors with one product still churn at 60%. And the mid-balance band (£50–125K) is where the bank loses the middle class: wealthier customers are retained noticeably better than moderately wealthy ones.

8. The Anomaly Audit

Unusual customers were flagged by four independent methods and cross-checked against each other. 1,054 customers (10.5% of the book) were flagged by at least one method; 186 by three or four.

Method	Flagged	Churn rate	What it catches
IQR (statistical)	374 (3.7%)	23.5%	Extreme single values (age, credit score)
Z-score (statistical)	141 (1.4%)	13.5%	The most extreme numeric tail
Isolation Forest (structural)	500 (5.0%)	49.0%	Odd combinations of attributes
DBSCAN noise (structural)	818 (8.2%)	52.2%	Customers far from every segment

The two structural methods agree strongly with each other (384 shared records) and both flag populations that churn at roughly 2.5× baseline — strangeness in this book is churn-shaped. Each of the 1,054 flagged customers was classified: **2 data errors** (impossible values, quarantined from business decisions), **583 rare-but-legitimate cases** (such as maximum-

product holders — monitor, never delete), and **469 risk signals** (churn-linked patterns warranting intervention, including the high-balance pre-churn group from Discovery 4).

9. What We Recommend, Monday Morning

Five actions follow directly from the evidence, ordered by expected value:

1. **Pre-emptive mid-life outreach.** Contact every 46–60-year-old, single-product customer within one quarter of going inactive. The 77% churn rate in this profile means a retention call here has four times the hit rate of a random one.
2. **German market review.** Treat 32.4% churn as a product-and-service problem, not a customer problem: benchmark German fees, rates and service against local competitors. Germany also supplies almost half the high-balance losses, so this review protects the most money per customer.
3. **Cross-sell as retention, not revenue.** Target second-product offers at single-product holders — the churn gap between one and two products (27.7% vs 7.6%) is the largest controllable lever in the book. Bundles aimed at the 46–60 band attack Discovery 1 and Discovery 3 at once.
4. **Relationship-manager alerts on anomalous high-balance accounts.** Wire the anomaly flags into account management: any account above the 75th balance percentile that picks up a structural-anomaly flag gets a human review. The 592 past losses in this profile averaged £149,755 each.
5. **Change the loyalty KPI.** Report relationship depth (products held, activity) alongside balance in management dashboards. Balance flatters exactly the segments most likely to leave.

10. Honest Limits of This Analysis

Four caveats keep these findings honest. The segments are working personas, not sharply separated populations — the statistical separation between clusters is modest, which is normal for behavioural banking data, and we validated the three-segment view across three different algorithms before trusting it. The rules describe association, not cause: being 46–60 does not make a customer leave, it marks the circumstances in which leaving happens, so the recommendations target the circumstances (inactivity, single products) rather than the demographics. The data is a single snapshot, so the “balance drop before closure” pattern in the brief can only be observed as its end state — high-balance accounts that closed — not as a movement over time. And the salary attribute carries almost no churn signal in this dataset; income bands appear in no strong rule, which is itself a finding: the bank loses customers by engagement, not by income.

11. The Central Question, Answered

What did we discover that was not already obvious from the raw data? **The raw data says one in five customers leaves. It does not say that departure is concentrated in a specific mid-life window (46–60) and multiplied by shallow engagement; that one market (Germany) drives double-rate churn across every segment simultaneously; that product depth — not balance — is what anchors a customer; or that the most expensive departures cluster among structurally unusual high-balance accounts that two independent detectors can flag in advance.** Each of those four sentences is invisible in a

column average, was found by a different mining technique, and carries a specific action. That is the difference between data and knowledge.

Appendix — Deliverables and Reproducibility

This report is one of three Phase 5 deliverables. The interactive dashboard (Python Dash) lives in the visualization folder: run “python prepare_data.py” once, then “python app.py”, and open <http://127.0.0.1:8050> — every control responds from memory in well under 100 ms. The 10-minute group presentation (Group5_Presentation.pptx) sits alongside this report. The complete technical record — preprocessing decisions, all three clustering algorithms with validation, the full Apriori run, and the four-method anomaly audit with cross-references — is the project notebook (notebooks/notebook.ipynb); every number in this report is reproducible from it with a fixed random seed.